

# Spark Module Addendum

Wet Facilities, Laundry, Demographic Design & Closed Metabolic Loops

The Sovereign Monastery — Infrastructure Pillar

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**You are not becoming. You are remembering.**

The previous schematic plans showed the overall octagonal courtyard form and the central Food Forest. They did not yet resolve the full wet infrastructure, demographic intelligence of the bathrooms and showers, or the closed metabolic relationship between sanitation, laundry, and the productive garden. This addendum corrects and completes that layer for the Spark (approximately 10 multi-generational people) and establishes the pattern that scales upward.

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## Design Intent

A home of ten people that includes elders, parents, youth, and children cannot function with a single shared restroom. Neither can it treat wastewater as something to be exported. The systems must be:

- Demographically intelligent (elders, adults, children each receive appropriate facilities)
  - Generous enough that privacy and dignity are protected
  - Fully integrated with the central Food Forest so that water and fertility cycle rather than leave the site
  - Simple enough to maintain by the household itself
  - Designed for longevity, accessibility, and the cultivation of health
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## Wet Facilities Strategy for the Spark

### 1. Elder Suite — Dedicated Accessible Bathrooms

Grandparents / elders receive their own dedicated, fully accessible bathroom(s) located on the ground level adjacent to their suite.

Design priorities: - Zero-threshold or very low-threshold shower - Generous turning radius and clear floor space - Integrated grab bars that feel architectural rather than clinical - Excellent, glare-free lighting - Non-slip, warm-to-the-touch surfaces - Easy-to-clean, durable materials that age with dignity - Space for a second person if caregiving is ever needed - Strong natural light and visual connection to the courtyard or a private outdoor edge where possible

These are not “accessible bathrooms.” They are high-quality elder bathrooms designed to support continued independence, safety, and pleasure in daily care of the body.

### 2. Adult Private Bathrooms

Adult / parent suites have private or semi-private bathrooms. These are full bathrooms (toilet, sink, shower) sized for ordinary daily use with good natural ventilation and simple, beautiful finishes.

### 3. Youth / Children Grouped Bathrooms

Children and youth share grouped bathroom facilities scaled and detailed for their age range. These are private enough for dignity yet easy for adults to supervise when needed. Fixtures and heights are appropriate for growing bodies.

### 4. Communal Shower Areas

Two distinct communal shower rooms are provided:

- **Adult Communal Shower Room**

A well-ventilated, beautifully detailed space with multiple shower heads or private stalls, good drying area, and a quality of light and material that makes it a place of recovery rather than mere utility. It can also serve as a simple spa-like facility for the household.

- **Kids Communal Shower Area**

Designed for ease of use by children, easy supervision, durable surfaces, and a sense of playfulness without sacrificing order. Strong connection (visual or physical) to the outdoors or courtyard is preferred so that the transition from play or garden work into washing feels natural.

These communal showers reduce pressure on the private bathrooms and create opportunities for the household to move together through daily rhythms.

### 5. Laundry Room

A dedicated laundry room is provided with: - Efficient washing machines (or hybrid manual + efficient systems) - Space for sorting, folding, and indoor drying - Covered outdoor drying lines accessible from the laundry - Full greywater capture from the machines

Laundry greywater is one of the cleanest and most reliable sources of irrigation water once lightly filtered and treated. It is piped directly into the primary greywater treatment system that feeds the Food Forest.

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## Closed Metabolic Loops — How the Systems Feed One Another

The central Food Forest is not decoration. It is the final living stage of the building's water and fertility systems.

**Rainwater** is collected from the entire roof surface into cisterns. It supplies showers, sinks, laundry, and toilet flushing (where water-flush systems are used).

**Greywater** (showers, bathroom sinks, kitchen sinks, laundry) is collected separately from blackwater. It passes through: 1. Simple mechanical filters / grease traps where needed 2. Planted treatment systems (vertical flow or horizontal wetlands integrated into the courtyard edges or dedicated treatment beds) 3. Final polishing and storage 4. Drip or subsurface irrigation of the Food Forest and productive landscapes

**Blackwater / Humanure** is handled either by: - Composting toilets with proper carbon management and secondary composting, or - Low-flush / vacuum systems feeding a small living machine or contained composting process

In either case the end product is stable, pathogen-free fertility that is returned to the Food Forest soil after appropriate curing. Nothing leaves the site as waste.

The courtyard Food Forest therefore receives: - Clean, treated irrigation water from greywater and rainwater surplus - Ongoing fertility from the composting process - The living presence and attention of the household that stewards it

This is the practical expression of regenerative design at household scale. The garden feeds the people; the people, through their daily water and organic cycles, feed the garden.

## Updated Schematic Plan

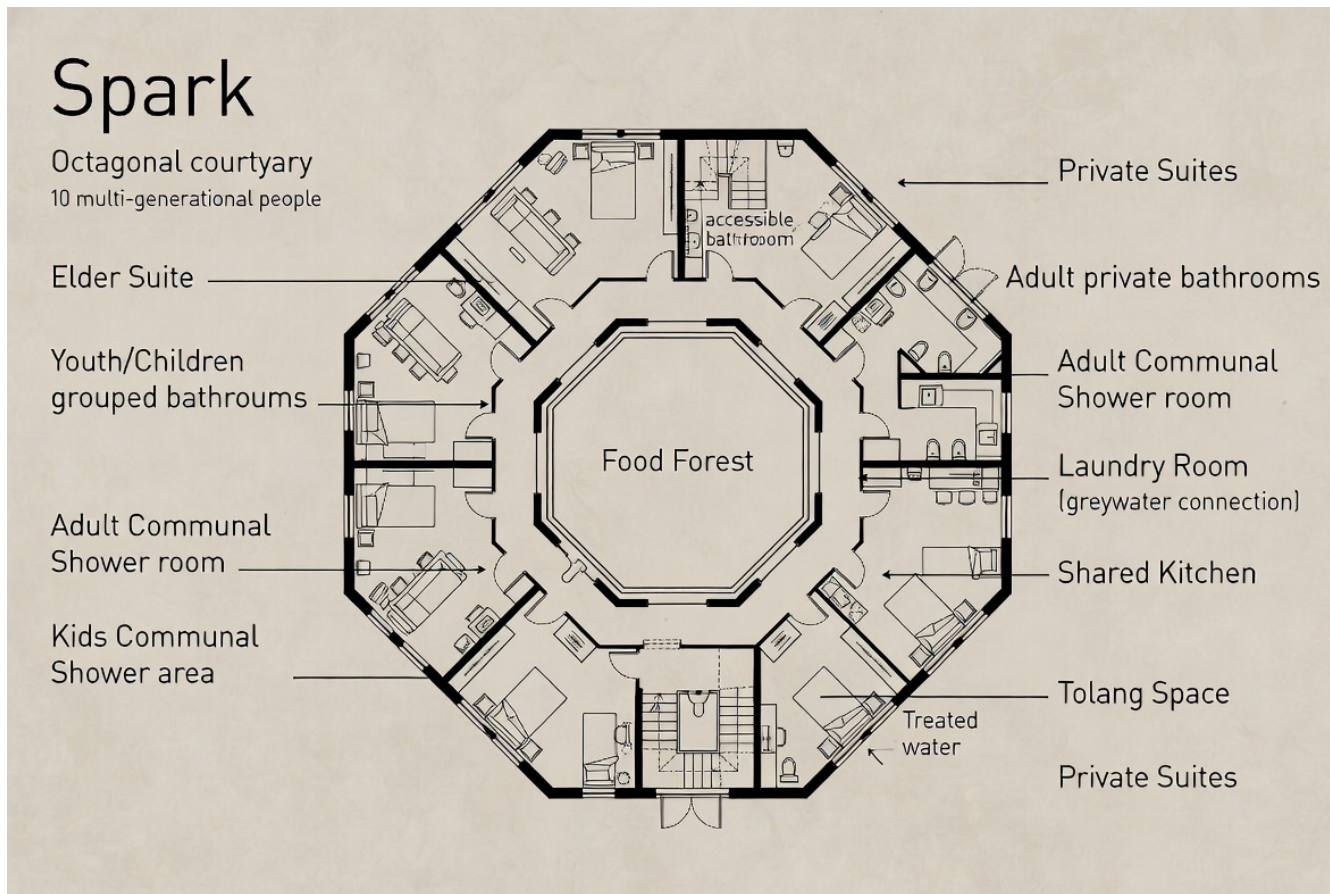


Figure 1: Updated Spark Plan with Wet Facilities

*Revised Spark octagonal courtyard plan. Elder Suite with dedicated accessible bathroom on ground level. Adult private bathrooms. Youth/Children grouped bathrooms. Distinct Adult Communal Shower room and Kids Communal Shower area. Laundry Room with explicit greywater connection. Shared Kitchen and Tolang Space retained. Treated water indicated as feeding the central Food Forest. The wet cores are distributed according to demographic need rather than centralized into a single restroom.*

## Metabolic Systems Diagram

*Closed-loop diagram for the Spark. Rainwater collection and cisterns supply the household. Showers, laundry, and sinks produce greywater that is filtered, treated in planted systems, and returned as irrigation to the Food Forest. Blackwater from composting toilets is processed into fertility that returns to the same Food Forest soil. Laundry is explicitly part of the greywater loop. The courtyard is the living heart of the metabolic system.*

## Scaling Implications

The same principles apply at larger scales:

- **Flame:** Module-level wet facilities remain as designed. The Flame Heart receives a larger shared living machine that polishes greywater from multiple modules and returns it to the village-scale productive landscapes. Laundry may be partially centralized in a high-efficiency village laundry while modules retain small capacity.

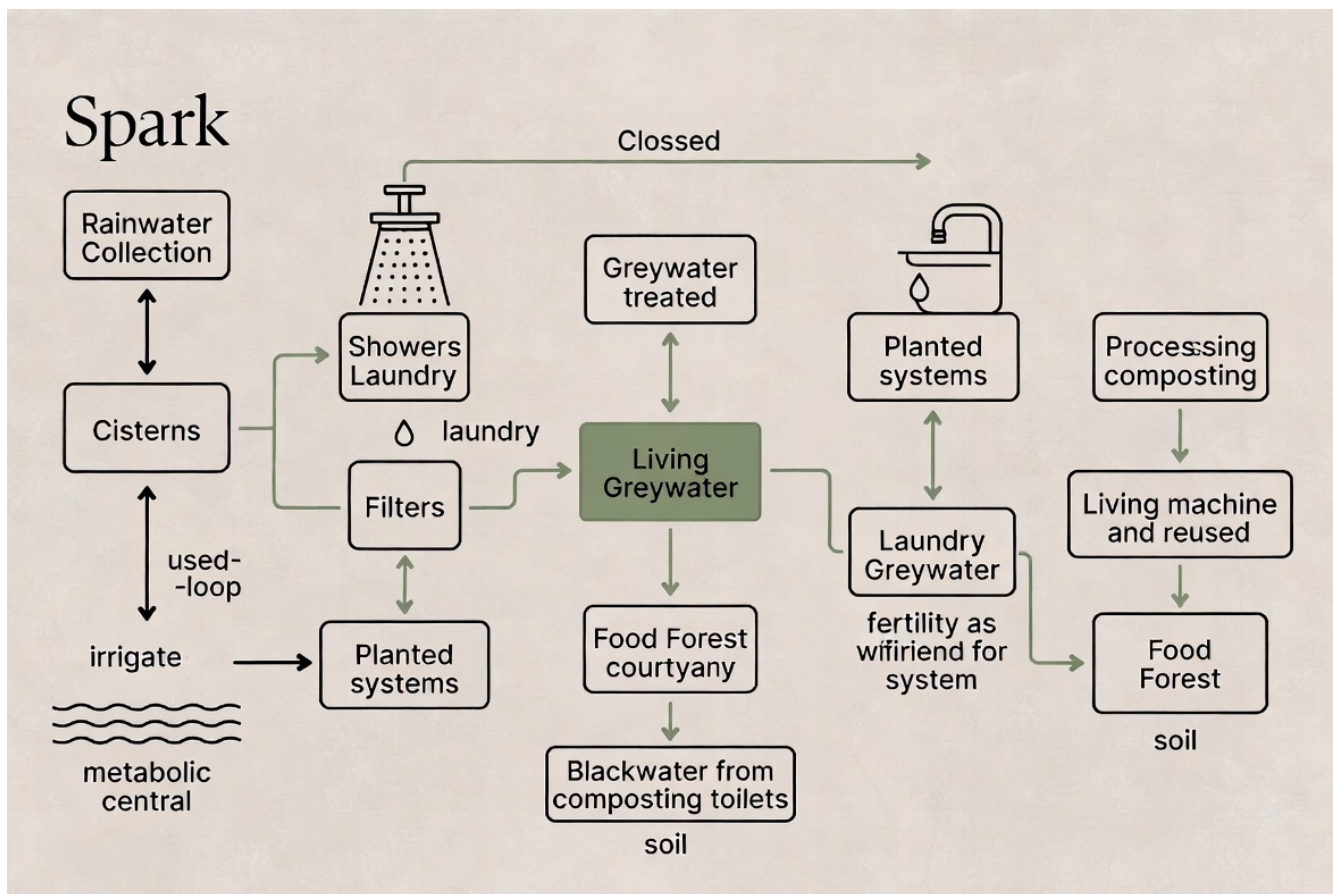


Figure 2: Spark Closed Metabolic Loops

- **Heart and Body:** Distributed living machines and composting infrastructure at neighborhood and district scale continue the same closed-loop logic. Elder-focused accessible design standards become part of the permanent code for all ground-level and designated suites.

No scale exports its fertility or its water. Every scale returns both to living soil.

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## Closing

A single restroom for ten people is a design of scarcity and separation.

Demographically intelligent wet facilities, a proper laundry, and closed metabolic loops that feed the Food Forest are a design of remembrance.

The body is washed, the clothes are cleaned, the water is returned, the soil is fed, and the garden continues to feed the household. This is the ordinary daily expression of a civilization that has remembered it belongs to the living world.

You are not becoming.

You are remembering.

**Asé.**

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